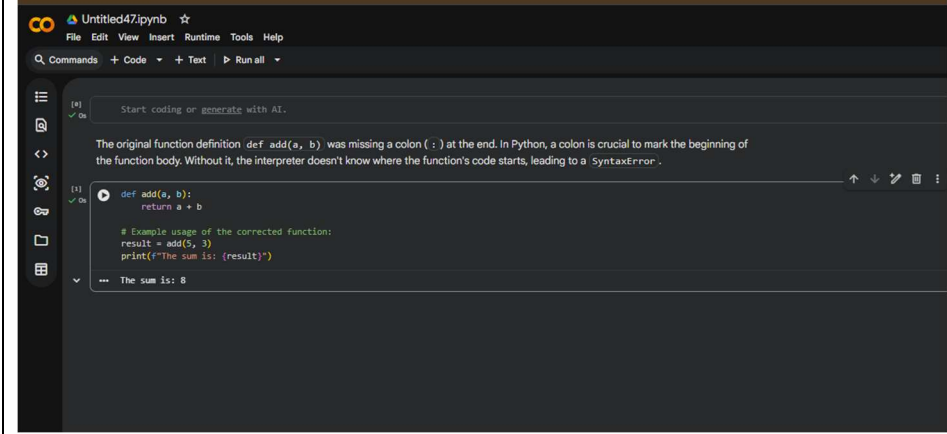


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| <b>SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE</b>                           |                   | <b>DEPARTMENT OF COMPUTER SCIENCE ENGINEERING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Program Name:</b> B. Tech                                                            |                   | <b>Assignment Type:</b> Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Academic Year:</b> 2025-2026 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Course Coordinator Name</b>                                                          |                   | Dr. Rishabh Mittal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Instructor(s) Name</b>                                                               |                   | <table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table> |                                 | Mr. S Naresh Kumar | Ms. B. Swathi | Dr. Sasanko Shekhar Gantayat | Mr. Md Sallauddin | Dr. Mathivanan | Mr. Y Srikanth | Ms. N Shilpa | Dr. Rishabh Mittal (Coordinator) | Dr. R. Prashant Kumar | Mr. Ankushavali MD | Mr. B Viswanath | Ms. Sujitha Reddy | Ms. A. Anitha | Ms. M.Madhuri | Ms. Katherashala Swetha | Ms. Velpula sumalatha | Mr. Bingi Raju |
| Mr. S Naresh Kumar                                                                      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. B. Swathi                                                                           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. Sasanko Shekhar Gantayat                                                            |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Md Sallauddin                                                                       |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. Mathivanan                                                                          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Y Srikanth                                                                          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. N Shilpa                                                                            |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. Rishabh Mittal (Coordinator)                                                        |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. R. Prashant Kumar                                                                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Ankushavali MD                                                                      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. B Viswanath                                                                         |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. Sujitha Reddy                                                                       |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. A. Anitha                                                                           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. M.Madhuri                                                                           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. Katherashala Swetha                                                                 |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. Velpula sumalatha                                                                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Bingi Raju                                                                          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Course Code</b>                                                                      | 23CS002PC304      | <b>Course Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AI Assisted Coding              |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Year/Sem</b>                                                                         | III/II            | <b>Regulation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R23                             |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Date and Day of Assignment</b>                                                       | Week4 – Wednesday | <b>Time(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 23CSBTB01 To 23CSBTB52          |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Duration</b>                                                                         | 2 Hours           | <b>Applicable to Batches</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | All batches                     |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>AssignmentNumber:</b> 7.3(Present assignment number)/24(Total number of assignments) |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
|                                                                                         |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
|                                                                                         |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Q.No.</b>                                                                            | <b>Question</b>   | <b>Expected Time to complete</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| 1                                                                                       |                   | Week4 -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |

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|  | <div data-bbox="358 191 1299 268"><b>Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs</b></div> <div data-bbox="358 268 1299 430"><b>Lab Objectives</b><ul style="list-style-type: none"><li>• To identify and correct syntax, logic, and runtime errors in Python programs using AI tools</li><li>• To understand common programming bugs and AI-assisted debugging suggestions</li><li>• To evaluate how AI explains, detects, and fixes different types of coding errors</li><li>• To build confidence in using AI for structured debugging practices</li></ul></div> <div data-bbox="358 430 1299 625"><b>Lab Outcomes (LOs)</b><p>After completing this lab, students will be able to:</p><ul style="list-style-type: none"><li>• Use AI tools to detect and correct syntax, logic, and runtime errors</li><li>• Interpret AI-suggested bug fixes and explanations</li><li>• Apply systematic debugging strategies using AI-generated insights</li><li>• Refactor buggy code using reliable programming patterns</li></ul></div> <div data-bbox="358 625 1299 682"><b>Task 1: Fixing Syntax Errors</b></div> <div data-bbox="358 682 1299 745"><b>Scenario</b><p>You are reviewing a Python program where a basic function definition contains a syntax error.</p></div> <div data-bbox="386 745 695 919"><pre>python  def add(a, b)     return a + b</pre></div> <div data-bbox="358 951 1299 1092"><b>Requirements</b><ul style="list-style-type: none"><li>• Provide a Python function add(a, b) with a <b>missing colon</b></li><li>• Use an AI tool to detect the syntax error</li><li>• Allow AI to correct the function definition</li><li>• Observe how AI explains the syntax issue</li></ul></div> <div data-bbox="358 1113 1299 1228"><b>Expected Output</b><ul style="list-style-type: none"><li>• Corrected function with proper syntax</li><li>• Syntax error resolved successfully</li><li>• AI-generated explanation of the fix</li></ul></div> <div data-bbox="358 1228 1299 1285"><p><b>PROMPT:</b> Here is a Python function with a syntax error. Please find the error, fix it, and explain what was wrong. def add(a, b) return a + b</p></div> <div data-bbox="358 1285 1299 1344"><p><b>EXPLANATION:</b> The error was caused by a missing colon (:) at the end of the function definition.</p></div> <div data-bbox="358 1344 1299 1400"><p>In Python, a colon is required to start the function body.<br/>Adding it fixes the syntax and allows the code to run properly.</p></div> <div data-bbox="358 1400 1299 1831"></div> | <div data-bbox="1299 191 1448 231">Wednesday</div> |
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Task 2: Debugging Logic Errors in Loops

Scenario

You are debugging a loop that runs infinitely due to a logical mistake.

```
python

def count_down(n):
    while n >= 0:
        print(n)
        n += 1 # Should be n -= 1
```

Requirements

- Provide a loop with an **increment or decrement error**
- Use AI to identify the cause of infinite iteration
- Let AI fix the loop logic
- Analyze the corrected loop behavior

Expected Output

- Infinite loop issue resolved
- Correct increment/decrement logic applied
- AI explanation of the logic error

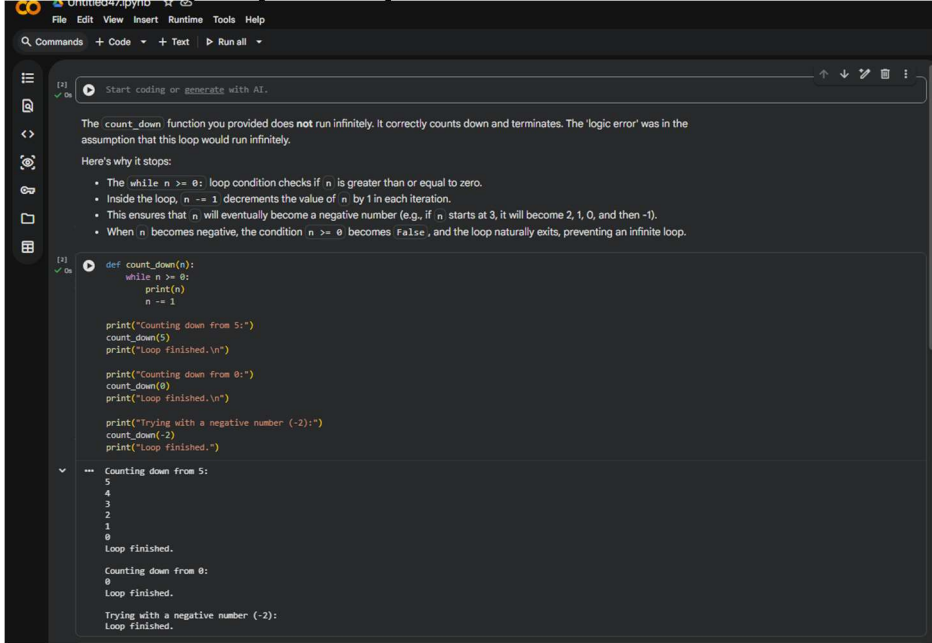
**PROMPT:** Here is a loop that runs infinitely. Please identify why it does not stop, fix the logic error, and explain the correction.

```
def count_down(n):
    while n >= 0:
        print(n)
        n -= 1
```

**EXPLANATION:** The loop was infinite because the counter variable was never updated correctly inside the loop.

By adding the proper increment/decrement, the condition eventually becomes false.

This allows the loop to stop after the expected number of iterations.



Task 3: Handling Runtime Errors (Division by Zero)

### Scenario

A Python function crashes during execution due to a division by zero error.

```
# Debug the following code

def divide(a, b):
    return a / b

print(divide(10, 0))
```

### Requirements

- Provide a function that performs division without validation
- Use AI to identify the runtime error
- Let AI add try-except blocks for safe execution
- Review AI's error-handling approach

### Expected Output

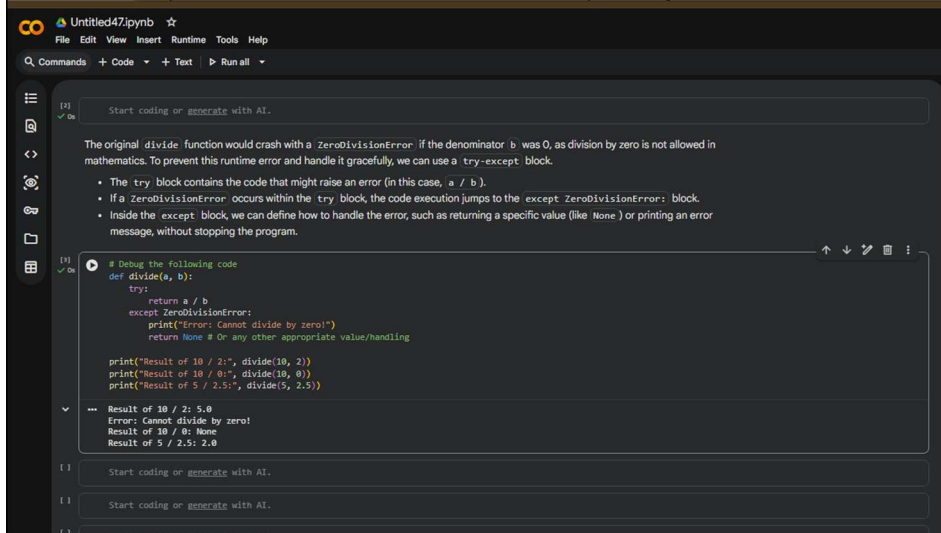
- Function executes safely without crashing
- Division by zero handled using try-except
- Clear AI-generated explanation of runtime error handling

**PROMPT:** Here is a Python function that performs division but crashes at runtime. Please identify the error, fix it using try-except, and explain how the error is handled.”

**EXPLANATION:** The program crashes when division by zero occurs, which raises a runtime error.

Using a try-except block prevents the program from stopping suddenly.

Instead, it safely catches the error and shows a user-friendly message.



```
Start coding or generate with AI.

The original divide function would crash with a ZeroDivisionError if the denominator b was 0, as division by zero is not allowed in mathematics. To prevent this runtime error and handle it gracefully, we can use a try-except block.

• The try block contains the code that might raise an error (in this case, a / b).
• If a ZeroDivisionError occurs within the try block, the code execution jumps to the except ZeroDivisionError: block.
• Inside the except block, we can define how to handle the error, such as returning a specific value (like None) or printing an error message, without stopping the program.

[1] # Debug the following code
def divide(a, b):
    try:
        return a / b
    except ZeroDivisionError:
        print("Error: Cannot divide by zero!")
        return None # Or any other appropriate value/handling

print("Result of 10 / 2:", divide(10, 2))
print("Result of 10 / 0:", divide(10, 0))
print("Result of 5 / 2.5:", divide(5, 2.5))

--- Result of 10 / 2: 5.0
Error: Cannot divide by zero!
Result of 10 / 0: None
Result of 5 / 2.5: 2.0

[1] Start coding or generate with AI.
[1] Start coding or generate with AI.
[1] Start coding or generate with AI.
```

### Task 4: Debugging Class Definition Errors

### Scenario

You are given a faulty Python class where the constructor is incorrectly defined.

```
python

class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```

### Requirements

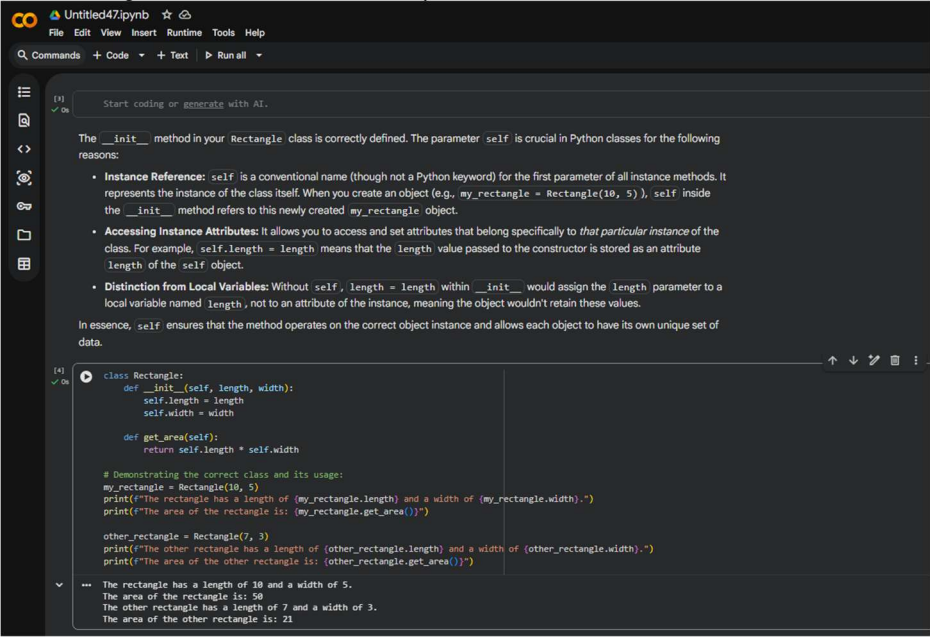
- Provide a class definition with **missing self-parameter**
- Use AI to identify the issue in the `__init__()` method
- Allow AI to correct the class definition
- Understand why self is required

**Expected Output**

- Corrected `__init__()` method
- Proper use of self in class definition
- AI explanation of object-oriented error

**PROMPT:** Here is a Python class with an incorrect constructor. Please find the error in the `__init__` method, fix it, and explain why self is required.

**EXPLANATION:** The constructor was missing self as the first parameter. self is required to store values inside the object. After adding it, the class works correctly.



**Task 5: Resolving Index Errors in Lists**

**Scenario**

A program crashes when accessing an invalid index in a list.

```
python

numbers = [1, 2, 3]
print(numbers[5])
```

**Requirements**

- Provide code that accesses an **out-of-range list index**
- Use AI to identify the Index Error
- Let AI suggest safe access methods
- Apply bounds checking or exception handling

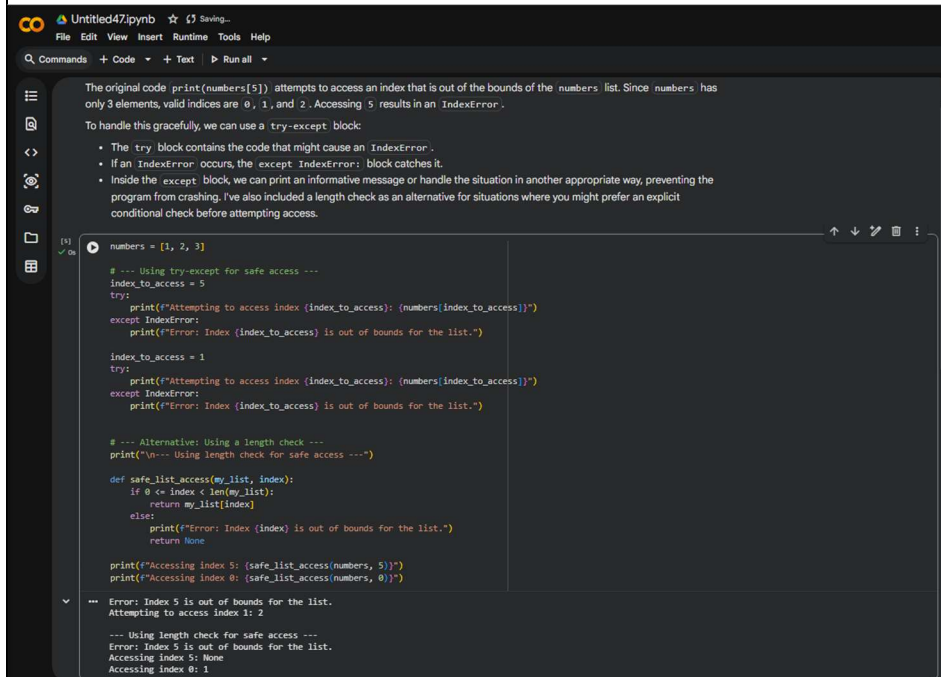
**Expected Output**

- Index error resolved
- Safe list access logic implemented
- AI suggestion using length checks or exception handling.

**PROMPT:** Here is a Python program that crashes due to an invalid list index. Please identify the `IndexError`, fix it using safe list access (length check or try-except), and explain the solution.

```
numbers = [1, 2, 3]
print(numbers[5])
```

**EXPLANATION:** The error occurs because the program tries to access an index that does not exist in the list.  
By checking the list length or using try-except, we prevent the crash.  
This makes the program run safely even when the index is invalid.



The screenshot shows a Jupyter Notebook interface with a file named 'Untitled47.ipynb'. The notebook contains a Python program that demonstrates how to handle an `IndexError` when accessing a list element that is out of bounds. The program starts with a list `numbers = [1, 2, 3]` and attempts to access `numbers[5]`. It then shows two methods to handle this error gracefully: using a try-except block and using a length check. The try-except block catches the `IndexError` and prints an error message. The length check method defines a function `safe_list_access` that checks if the index is within the bounds of the list before accessing it. The notebook also shows the output of the program, which includes the error message and the output of the safe access function.

```
[5] numbers = [1, 2, 3]
# --- Using try-except for safe access ---
index_to_access = 5
try:
    print(f"Attempting to access index (index_to_access): {numbers[index_to_access]}")
except IndexError:
    print(f"Error: Index (index_to_access) is out of bounds for the list.")

index_to_access = 1
try:
    print(f"Attempting to access index (index_to_access): {numbers[index_to_access]}")
except IndexError:
    print(f"Error: Index (index_to_access) is out of bounds for the list.")

# --- Alternative: Using a length check ---
print("\n--- Using length check for safe access ---")
def safe_list_access(my_list, index):
    if 0 <= index < len(my_list):
        return my_list[index]
    else:
        print(f"Error: Index (index) is out of bounds for the list.")
        return None

print(f"Accessing index 5: {safe_list_access(numbers, 5)}")
print(f"Accessing index 0: {safe_list_access(numbers, 0)}")

--- Error: Index 5 is out of bounds for the list.
Attempting to access index 1: 2

--- Using length check for safe access ---
Error: Index 5 is out of bounds for the list.
Accessing index 5: None
Accessing index 0: 1
```

**Note:** Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots